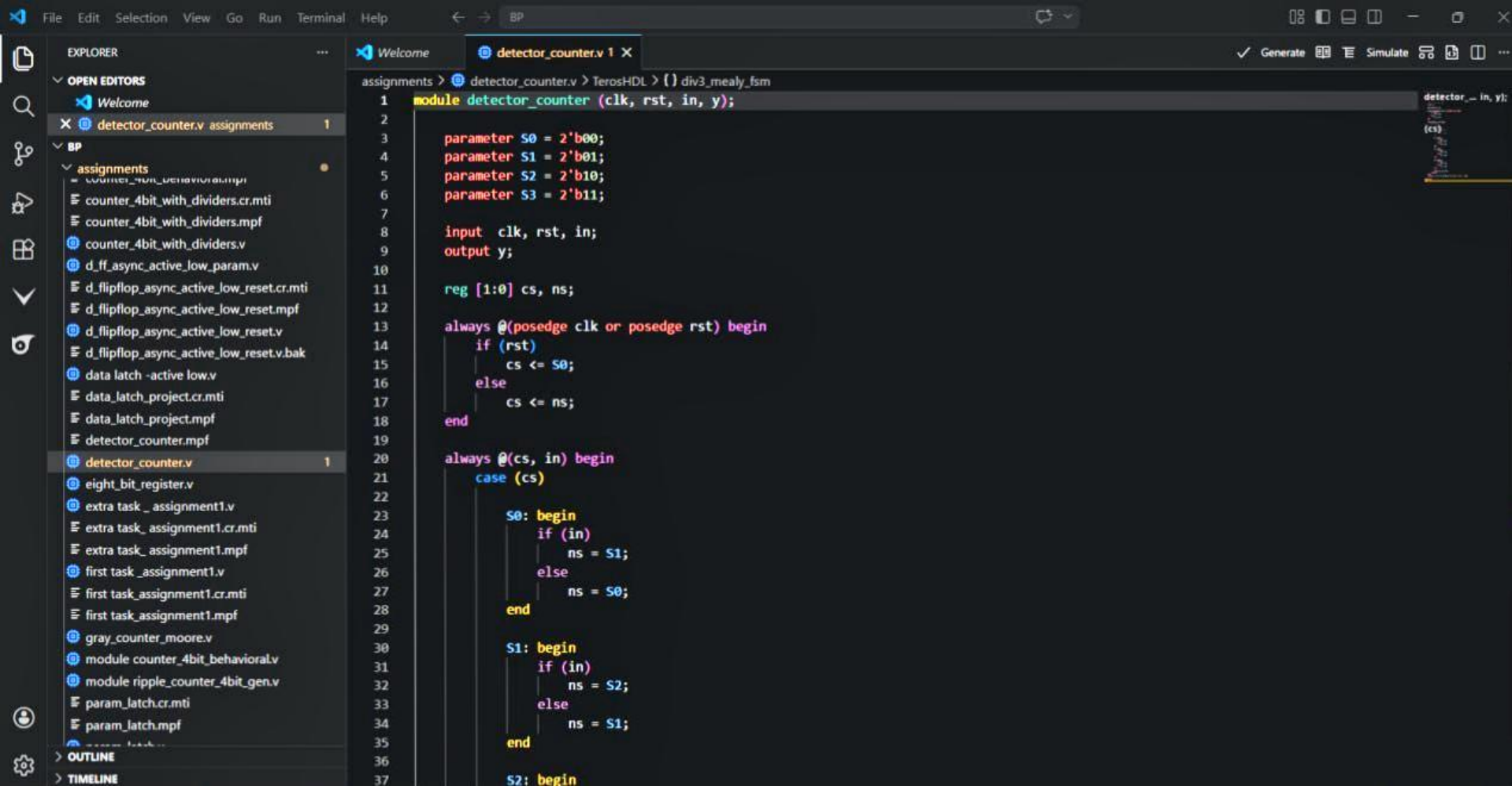




Welcome

detector_counter.v 1 X

Generate Simulate

assignments > detector_counter.v > TerosHDL > div3_mealy_fsm

```
1 module detector_counter (clk, rst, in, y);
```

```
2
3
4 parameter S0 = 2'b00;
5 parameter S1 = 2'b01;
6 parameter S2 = 2'b10;
7 parameter S3 = 2'b11;
```

```
8
9 input clk, rst, in;
10 output y;
```

```
11
12 reg [1:0] cs, ns;
```

```
13 always @(posedge clk or posedge rst) begin
14     if (rst)
15         cs <= S0;
16     else
17         cs <= ns;
18 end
```

```
19
20 always @(cs, in) begin
21     case (cs)
```

```
22
23         S0: begin
24             if (in)
25                 ns = S1;
26             else
27                 ns = S0;
28         end
```

```
29
30         S1: begin
31             if (in)
32                 ns = S2;
33             else
34                 ns = S1;
35         end
```

```
36
37         S2: begin
```

detector_... in, y);

(cs)

assignments >  detector_counter.v > TerosHDL > { } div3_mealy_fsm

```

1  module detector_counter (clk, rst, in, y);
36
37      S2: begin
38          if (in)
39              ns = S3;
40          else
41              ns = S2;
42      end
43
44      S3: begin
45          if (in)
46              ns = S1;
47          else
48              ns = S0;
49      end
50
51      default: ns = S0;
52  endcase
53  end
54  assign y = (cs == S2 && in == 1'b1) ? 1'b1 : 1'b0;
55
56  endmodule
57  ~~~

```

```
detector_... in, y);  
// ...  
(cs)  
// ...
```

